PACKAGE INSERT

Product Name: BEACON-FDG MULTIDOSE INJECTION

Content: 18F-Fluorodeoxyglucose

Package Insert Colour: Black and White

Name & Address of the Manufacturer: Beacon Hospital Sdn. Bhd.
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Composition

In 1 ml of FDG injection contains
^{18}F-Fluorodeoxyglucose (2-deoxy-2-[18F] fluoro-D-glucose) - 185-7400 MBq
(5-200 mCi) per ml

General

F-Fluorodeoxyglucose Injection is a radiolabel analogue of glucose that is rapidly distributed to all organs of the body after intravenous administration. After background clearance of ^{18}F-Fluorodeoxyglucose, optimal PET imaging is often achieved between 30 to 40 minutes after administration.

Shelf Life

The shelf life of ^{18}F-Fluorodeoxyglucose is 10 hours, based on stability studies conducted in our facility, Beacon International Specialist Centre (Formerly known as Wijaya International Medical Centre Sdn Bhd.).

In 4 healthy male volunteers, receiving an intravenous administration of 30 seconds in duration, the arterial blood level profile for ^{18}F-Fluorodeoxyglucose was described as a triexponential decay curve. The effective half-life ranges of the three phases were 0.2-0.3 minutes, 10-13 minutes with a mean and standard deviation (STD) of 11.6 ± 1.1 min, and 80-95 minutes with a mean and STD of 88 ± 4 min. Within 33 minutes, a mean of 3.9% of the administration radioactive dose was measured in the urine. The amount of radiation exposure of the urinary bladder at two hours post-administration suggests that 20.6% (mean) of the radioactive dose was present in the bladder. See the Metabolism Section for additional clearance times.

Absorption

FDG acts as a glucose analogue in the body and is transported from the plasma into the cells by glucose transporters (most important are GLUT 1 and GLUT 4). The increased glucose uptake in cancer cells is due to increased anaerobic glycolysis, [1],[2] a phenomenon known as the " *Warburg Effect*" and is postulated to be due to up regulation of glucose transporters and hexokinase levels and lower levels of the enzyme glucose-6- phosphatase in cancer cells. Once it is absorbed into the cells, FDG is trapped in the cell until F-18 is decayed to O-18.

Distribution

FDG is an analog of glucose and is used as a tracer of glucose metabolism. Therefore, the distribution of FDG is not limited to malignant tissue. Myocardial uptake is highly variable ranging from very intense to nil. Salivary gland and tonsillar uptakes are commonly seen and are somewhat variable. The lungs typically show very low uptake with somewhat higher uptake in the mediastinum. The liver shows modest uptake (usually slightly higher than mediastinum) with somewhat less intense uptake seen in the spleen. Uptake in the gastrointestinal (GI) tract (esophagus, stomach, colon) is highly variable. Since FDG is excreted by the kidneys, there is frequently intense FDG activity in the renal collecting system and urinary bladder, with less activity in the renal cortex. Muscle shows low uptake at rest. Mild activity is seen in hematopoietic bone marrow but no bone uptake is present. Thyroid uptake is usually minimal and non-specific. Breast uptake is variable with mild to moderate. glandular uptake commonly seen in pre menopausal women and with estrogen therapy. Relatively intense uptake is seen in lactating breasts, but the uptake is largely glandular with little activity in breast milk. Generally, low breast uptake is present in the postmenopausal state.

Metabolism

^{18}F-Fluorodeoxyglucose is transported into cells and phosphorylated to [^{18} F] – FDG-6-phosphate at a rate proportional to the rate of glucose utilization within that tissue. [^{18} F] – FDG-6-phosphate presumably is metabolized to 2-deoxy-2-[^{18} F] fluoro-6-phospho-D-mannose ([^{18} F] FDM-6-phosphate). ^{18}F-Fluorodeoxyglucose Injection may contain several impurities, e.g., 2-deoxy-2-chloro-D-glucose (CIDG). Bio distribution and metabolism of CIDG are presumed to be similar to ^{18}F-Fluorodeoxyglucose, and would be expected to result in intracellular formation of 2-deoxy-2-chloro-6-phospho-D-glucose (CIDG-6-phosphate) and 2-deoxy-2-chloro-6-phospho-D-mannose (CIDM-6-phosphate). The phosphorylated deoxy glucose compounds are dephosphorylated, and the resulting compounds, (FDG, FDM, CIDG and CIDM) presumably leave cells by passive diffusion. FDG and related compounds are cleared from non-cardiac tissues within 3 to 24 hours after administration. Clearance from the cardiac tissue may require more than 96 hours. ^{18}F-Fluorodeoxyglucose that is not involved in glucose metabolism in any tissue is excreted unchanged in the urine.

Urinary Excretion

^{18}F-Fluorodeoxyglucose is cleared from most tissues within 24 hours, and can be eliminated from the body unchanged in the urine. Three elimination phases have been identified in the reviewed literature. A early elimination phase (0.2-0.3 minutes) and two late phases (11.6 ± 1.1 min and 88 ± 4 min or 4.21 ± 1.09 and 50.08 ± 14.62). Within 33 minutes, a mean of 3.9% of the injected dose was measured in the urine of normal subjects. Within two hours of administration, a mean of 20.6% of the injected dose was found in the bladder.

Plasma Protein Binding: The extent of binding of ^{18}F-Fluorodeoxyglucose to plasma proteins is not known.

Pharmacokinetics in Special Populations
Extensive dose range and dose adjustment studies with this drug product in normal and special populations have not been completed. In paediatric patients with epilepsy, doses given have been as low as 2.6 mCi.
The pharmacokinetics of ^{18}F-Fluorodeoxyglucose in renally impaired patients has not been characterized. ^{18}F-Fluorodeoxyglucose is eliminated through the renal system. Care should be taken to prevent excessive and unnecessary radiation exposure to this organ system and adjacent tissues.
The effects of fasting, varying blood sugar levels, conditions of glucose intolerance, and diabetes mellitus on ^{18}F-Fluorodeoxyglucose distribution in humans have not been ascertained. Diabetic patients may need stabilization of blood glucose levels on the day before and on the day of the ^{18}F-FDG study.

Pharmacodynamics
^{18}F-Fluorodeoxyglucose is a glucose analog that concentrated in cells that rely upon glucose as an energy source, or in cells whose dependence on glucose increases under pathophysiological conditions. ^{18}F-Fluorodeoxyglucose is transported through the cell membrane by facilitative glucose transported proteins and is phosphorylated within the cell to [^{18}F] FDG-6-phosphate by the enzyme hexokinase. Once phosphorylated, it cannot exit until dephosphorylated by glucose-6-phosphatase. Therefore, within a given tissue of pathophysiological process, the retention and clearance of ^{18}F-FDG reflect a balance involving glucose transporters, hexokinase and glucose-6-phosphatase activities. When allowance is made for the kinetic differences between glucose and FDG transport and phosphorylation (expressed as the "lumped constant" ration), ^{18}F-Fluorodeoxyglucose is used to assess glucose metabolism.
In comparison to background activity of the specific organ or tissue type, regions of decreased or absent uptake of ^{18}F-Fluorodeoxyglucose reflect the decrease or absence of glucose metabolism. Regions of increased uptake of ^{18}F-Fluorodeoxyglucose reflect greater than normal rates of glucose metabolism.
In cancer, the cells are generally characterized by enhanced glucose metabolism. Regions of increased uptake of ^{18}F-Fluorodeoxyglucose reflect greater than normal rates of glucose metabolism.
In cancer, the cells are generally characterized by enhanced glucose metabolism partially due to (1) an increase in the activity of glucose transporters, (2) an increase rate of phosphorylation activity, (3) a reduction of phosphatase activity or, (4) a dynamic alteration in the balance among all these processes. However, glucose metabolism of cancer as reflected by ^{18}F-Fluorodeoxyglucose accumulation may be increased, normal or decreased. Also, inflammatory cells can have the same variability of uptake of ^{18}F-Fluorodeoxyglucose.
In the heart under normal aerobic conditions, the myocardium meets the bulk of its energy requirements by oxidizing free fatty acids. Most of the exogenous glucose taken up by the myocyte is converted into glycogen. However, under ischemic conditions; the oxidation of free fatty acids decreases, exogenous glucose becomes the preferred myocardial substrate, glycolysis is stimulated, and glucose taken up by the myocyte is metabolized immediately instead of being converted into glycogen. Under these conditions, phosphorylated ^{18}F-Fluorodeoxyglucose accumulates in the myocyte and can be detected with PET imaging.
Normally, the brain relies on anaerobic metabolism. In epilepsy, the glucose metabolism varies. Generally during a seizure, glucose metabolism increases. Intricately, the seizure focus tends to by hypo metabolic.

Indication
^{18}F-Fluorodeoxyglucose (2-deoxy-2-[18F] fluoro-D-glucose) is used as contrast medium in PET (Positron Emission Tomography) for the identification of regions of abnormal glucose metabolism associated with foci of epileptic seizures and tumour.
Dosage: Adult dose is between 5-13 mCi and children are 0.21 mCi/Kg.
Contra Indication: Pregnancy and lactation.
Special Precaution: The patient should be rest and keep isolated from anybody for at least 1 hour and after FDG injection.
Adverse Reaction: No data available.
Drug-Drug Interactions: Drug-Drug Interactions with ^{18}F-Fluorodeoxyglucose have not been evaluated.
Date of last revision: 28th December 2023.